

LANE A — confinement property tests: RESULTS (L1 CONFIRMS, L2 REFUTES — reported beside)

Two pre-registered membership questions, run as property tests, never mass computations (mass gate closed). One confirms, one refutes — and the refutation is the more useful result. Reconnected first (grep the proved theorem): T1930 (color = short-root spaces, $m_s = N_c = 3$), F727 (charged leptons color-singlet, $V_{12}=0$), T2517 (generation addresses $\{5/2, 3/2, 0\}$ radial), round-4 Šilov ($\lambda_2 > 0 \Rightarrow$ zero boundary), F603/OP-6 (rank-1 (1,0) spherical = Higgs).

L1 — is the color block V_{12} at $\lambda_2 > 0$? CONFIRMED (DERIVED-structure).

The membership chain, each step banked: - color lives in the short-root spaces (T1930): short-root multiplicity $m_s = N_c = 3$; the three short-root directions per short root are the $SU(3)$ -color fundamental. The color $SO(3) = SO(V_{12})$ acts on the short-root space. - the short-root axis is λ_2 (round-4 / B_2): a K-type (λ_1, λ_2) carries short-root weight $\Leftrightarrow \lambda_2 > 0$. - therefore: colored $\Leftrightarrow \lambda_2 > 0 \Leftrightarrow$ confined (round-4: $\lambda_2 > 0 \Rightarrow$ zero Šilov value). Color-nonsinglet modes carry short-root weight, so they never reach the boundary.

$\Rightarrow$ The confinement projection's kernel IS the colored sector. Color confinement is a *reading of one projection* $\Pi_{\check{S}}$ on D_{IV}^5 : colored = short-root ($\lambda_2 > 0$) = confined = zero Šilov value; color-singlet ($\lambda_2 = 0$) = emitted = reaches the boundary.

Two independent cross-checks (both banked, both agree): - charged leptons are color-singlets (F727, $V_{12}=0$, live in the frame $\mathbb{R}c_1 \oplus \mathbb{R}c_2 \rightarrow \lambda_2 = 0 \rightarrow$ emitted (leptons are free/observable, not confined). yes - the Higgs condensate = rank-1 $O = SO(5)$ vector (1,0) (F603/OP-6) $\rightarrow \lambda_2 = 0 \rightarrow$ spherical, boundary-reaching (an emitted, observable condensate). yes

Tier: DERIVED-structure. The identification “confinement kernel = colored sector” composes two banked theorems (T1930 + round-4); it is a *property* statement (which modes are confined), not a spectrum/mass claim. Target-innocent. This is the geometric statement of color confinement, and it survives its cross-checks.

L2 — is the $\nu=0$ generation in the confinement kernel? REFUTED — and the refutation clarifies.

Pre-registered form: “ $m_1=0 \Leftrightarrow$ the $\nu=0$ generation $\in \ker \Pi_{\check{S}} (\lambda_2 > 0)$.” Run: - factual correction first: $\nu=0$ is the tau (Šilov-point address, T2517), the *heaviest* charged lepton — not the lightest (I mis-stated it as lightest in the pre-registration; owned). - the tau is a charged lepton $\Rightarrow$ color-singlet (F727) $\Rightarrow \lambda_2 = 0 \Rightarrow$ emitted, *not* in the $\lambda_2 > 0$ kernel.

$\Rightarrow$ L2 REFUTES: the $\nu=0$ generation is emitted, not confined. And the reason is structural and worth banking: the confinement axis (λ_2 , short-root / color) is *or-*

thogonal to the generation / mass axis (ν , radial / Korányi–Wolf stratum). My lead conflated two different coordinates: - λ_2 = the SO(5) short-root weight (angular, color) — sets *confinement*; - ν = the radial boundary-orbit address $\{5/2, 3/2, 0\}$ (the Cartan-slice / KW stratum) — sets the *generation*.

A charged lepton runs down the ν -axis at fixed $\lambda_2=0$ (always color-singlet, always emitted). So the “ $m_1=0$ engine” is a *radial* (ν) rank-1 phenomenon — the rank-1 condensate (1,0), itself $\lambda_2=0$ — and is NOT the λ_2 -confinement kernel. The corpus caption “confinement, $m_1=0$ engine” bundles two distinct mechanisms on two distinct axes; this test separates them.

Tier: REFUTED lead $\rightarrow$ a structural clarification (two orthogonal axes). No mass computed. Reported beside L1 (C6 discipline: the miss beside the hit).

What promotes, what the two axes are

reading	axis	tier
color confinement = ker $\Pi_{\check{S}}(\lambda_2>0 = \text{colored})$	λ_2 (short-root / color)	DERIVED-structure (L1)
generation / $m_1=0$ rank-1 condensate	ν (radial / KW stratum)	separate (mass gate closed)
the two axes are orthogonal	—	DERIVED (L2 refutation — a clarifying result)

Ledger update (this note)

- A-L1 $\rightarrow$ HIT (Derived-structure): colored $\Leftrightarrow \lambda_2>0 \Leftrightarrow$ confined; leptons (color-singlet, $\lambda_2=0$) and Higgs (1,0) emitted — cross-checks pass.
- A-L2 $\rightarrow$ MISS (refuted, reported): $\nu=0$ (tau) is color-singlet $\rightarrow$ emitted, not confined; confinement (λ_2) perp generation (ν). Owned the “lightest” misstatement.

Lyra, 2026-08-21 (Lane A, confinement property tests). L1 CONFIRMED at DERIVED-structure: color confinement = the round-4 projection kernel — colored $\Leftrightarrow$ short-root $\Leftrightarrow \lambda_2>0 \Leftrightarrow$ zero Šilov value $\Leftrightarrow$ confined (T1930 + round-4), with two banked cross-checks (leptons color-singlet emitted; Higgs (1,0) emitted). L2 REFUTED: $\nu=0$ is the tau (color-singlet, $\lambda_2=0$, emitted) — NOT in the kernel; the confinement axis λ_2 is ORTHOGONAL to the generation/mass axis ν ; owned the mis-stated ‘lightest.’ The refutation is the useful result — it separates two mechanisms the corpus bundled. Membership tests only, no masses. Nothing pushed; CP existence-only.